Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Generative Audio, Music ML, Signal Processing | PhD

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

nishal.silva@hotmail.com | +1 613 200 2030 | Toronto, ON | nishal.xyz | linkedin.com/in/nishal-silva

---

5th November 2026

Re: Research Scientist, Generative Audio - Spotify (Remote (North America))

Dear Hiring Manager,

I am applying for the Research Scientist, Generative Audio role at Spotify. I am a machine learning researcher with a PhD from the University of Trento (2025), 10+ years across industry and academia, and a long-standing focus on music and audio ML (I am also a Trinity Grade 8 guitarist). My most recent published system made real-time decisions in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026).

Here is how my work maps to the role:

- **Generative audio.** I am building VarianceEngine, which generates expressive musical variations from a single performance recording. I evaluated 5 candidate architectures (mel-diffusion, VQ-VAE with transformer prior, retrieval, symbolic transcription, foundation-model fine-tuning) before choosing LoRA fine-tuning of MusicGen-medium with a custom audio-to-audio conditioning head. At McGill I explored diffusion- and VAE-based synthetic data generation.
- **Audio and signal processing at scale.** I released four open-access musical audio datasets (9,800+ recordings, 70+ musicians) and the C++17 library Nebula for audio feature extraction.
- **Rigorous experiments and publication.** I benchmark against honest baselines and publish what holds up: 10+ peer-reviewed papers (JAES, IEEE), and 6 methods over 100,000 trials in my latest benchmark.

To be direct about the gaps: my publications are at JAES and IEEE venues, not ICASSP, ISMIR, NeurIPS or ICML, and I have not yet done preference alignment or reward-model work. VarianceEngine is in progress, not a shipped system. I would start from the editing and conditioning side, where my music-signal background is strongest.

I am based in Toronto and a Canadian permanent resident, eligible to work in Canada without sponsorship, and I understand the posting asks for North American candidates near a Spotify work location. I would welcome the opportunity to discuss how my work can support the Artist-First AI Music Lab.

Sincerely,

Nishal Stanislaus Silva, PhD